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MDCAT PAST PAPER

2022

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Medicoengineer.com

Dow University of Health Sciences Karachi

Examinations Department

Medical Universities of Sindh

MDCAT-Test 2022

Time Allowed: Three Hours and 30 Minutes

BIOLOGY

1) Hepatitis is caused by:
a) Protozoans
b) Algae
c) Bacteria
d) Fungi
e) Virus

2) Syphilis is caused by a bacteria called as:
a) Bacilli
b) Trichomonas
c) Neisseria gonorrhea
d) Genital herpes
e) Treponema palladium

3) Presence of remnant of pelvic girdle in whale and snake although they never have legs, shows evidence from:
a) Biochemical evidence
b) Paleontology
c) Vestigial organs
d) Embryology
e) Taxonomy

4) Photosynthetic bacteria liberate:
a)CO_2
b) H_2S
c)Sulphur
d) Hydrogen
e) O_2

5) Plasma membrane is:
a) Permeable
b) Selective permeable
c) impermeable
d) Sub permeable
e) Non permeable

6) Water is the most abundant compound in living organism its amount varies from:
a) 30% to 60%
b) 50% to 60%
c) 10% to 20%
d) 70% to 90%
e) 40% to 50%

7) The phase in which bicuspid and tricuspid valves open & semilunar valve close which produce 2nd heart sound (dub or dup). This the phase is known as:
a) Atrial diastole
b) Ventricular diastole
c) Ventricular systole
d) Contraction of heart
e) Systole

8) Phenotypic ratio of law of independent assortment of genes:
a) 9:7
b) 13:3
e) 9:3:3:1
d) 15:1
e) 12:3:1

9) Triose and glycerol is used in the synthesis of:
a) Carbohydrate
b) Protein
c) Lipid
d) Fatty acid
e) Phospholipid

10) They are inhibitors of key enzymes in the nervous system:
a) Calcium and Potassium
b) Substrate and activators
c) Hexokinase and glucose

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d) Magnesium and manganese
e) Pesticides DDT and parathione

11) Plant hormone associated with apical dominance is:
a) Auxin b) Gibberellin
c) Cytokinin d) Ethene e) Abscisic acid
12) Bolus enters the oesophagus not the traches due to presence of:
a) Roof palate b) Mouth c) Epiglottis d) Buccal cavity e) Tongue
13) Net ATP produced at the end of glycolysis:
a) 2 ATP b) 4 ATP
c) 6 ATP d) 3 ATP e) 8ATP
14)Class turbellaria, Trematoda, Cestoda belongs to phylum:
a) Echinodermata b) Porifera
c) Aschelminthes d) Cnidaria
e) Platyhelminthes
15) Spikelet inflorescence is the characteristic of family:
a) Solanaceae b) Rosaceae c) Fabceae d) Poaceae e) Mimosaceae
16) Each myosin is surrounded by:
a) Motor nerves b) Myosin head
c) 6-actin filaments d) filaments
e) 4-actin filaments
17) Synthesis of disaccharide by condensation two glucose joined by glycosidic linkage to form:
a) Galactose b) Lactose c) Fructose d) Phosphate e) Maltose
18) White eyed trait in drosophila is due to:
a) Linkage b) Mutati c) Crossing over d) Genome e) Gene pool

19) Enzymes are generally inactivated rapidly by exposure to UV high and also alpha, B rays because it alters the:
a) Quantity of enzyme b) pH of enzyme c) Shape of enzymes
d) Temperature of enzyme e)Nature of enzyme
20)In frog New air drawn into bucco-pharyngeal cavity through nostrils and old air retained under pressure in:
a) Lungs b) Nasal c) Glottis d) Trachea e) Nostrils
21) Genetic information in DNA is stored in the form of
a) Genetic code b) Chromosome c) Amino acid d) Helix e) Hydrogen bond
22) Pyruvic acid is acid of:
a) 4 carbon compound b) 5 carbon compound c) 6 carbon compound d) 7 carbon compound e) 3 carbon compound
23) Plant cell wall is made up of:
a) Lignin b) Hemicellulose
c) Putin d) Chitin e) Cellulose
24) Plant like character found in Euglena:
a) Pyrenoid b) Cilia c) Photoreceptor d) Flagellum
e) Pellicle
25) It has neurosecretory cells that secrete releasing & inhibiting hormone:
a) Pituitary gland b) Thyroid gland
c) Adrenal gland d) Master glands
e)Hypothalamus
26) It is the seat of all conscious activity:
a) Hind brain b) Cortex c) Spinal cord d) Mid brain
e) Medulla oblongata
27) The common energy currency of the cell is:
a) NHDP b) ADP c) ATP d) NADH e) PEPs
28) Scorpoid cyme is found in:

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a) Cotton b) Mulberry c) Sunflower d) Sundew e) Banana

29) Bacteria having flagella all over the surface:

a) Lophotrichous b) Peritrichous c) Atrichous d) Monotrichous e)Diplotrichous

30) A hormone secreted from the duodenal mucosa that changes the inactive trypsinogen into trypsin is:

a) Peptone b) Ecdysone c) Enterokinase d) Amylase e)Protease

31) Cell become dead due to the deposition of:

a) Cellulose b) Cutin c) Chitin d) Lignin e) Hemicellulose

32)Rabies is caused by:

a) Paramyxovirus b) Rhinovirus c) Arbovirus d) Rhabdovirus e) Poliovirus

33) In AIDS "D" stands for:

a) Different b) Difficult c) Deficiency d) Degenerate e) Decompose

34)It is a loop or set or response shown by the effectors that weaken the affect of the initial stimuli:

a) Negative feed back b) Neutral condition c) Positive feed back d) Static state e)Thermal state

35) Crossing over in meiosis occurs during the following stage

a) Diakinesis b) Diplotene c) Leptotene d) Zygotene e) Pachytene

36) A co-enzyme constitute about certain % of portion of entire enzyme:

a) 5% b) 1% c)2% d) 3% e)4%

37) Each chain of DNA is formed by the union of many nucleotides by means of:

a) Double bond b) Phosphodiester bonds c) Covalent bond d) Hydrogen bond e)Single bond

38) Mimosa pudica shows:

a) Thermonastic b) Haptonastic c) Nyctinastic d) Seismonastic e) Photonastic

39) Mendel's for his research selected:

a) Edible-pea b) Garden pea c) Bean plant d) Wild pea e) Pea-nut

40) This lipid is most common abundant richest source of energy in plant, and animal body:

a) Phospholipid b) Carotenoids c) Terpenoids d) Terpenes e) Acylglycerol

41) The organs perform same functions in different groups of organism but are different in this internal structure called as:

a) Homologous organs b) Bisymmetry organs c)Analogous Organ d) Radial symmetry e) Asymmetry

42) The hard parts of the seeds is, due to the presence of:

a) Collenchyma b) Sclerenchyma c) Parenchyma d) Fibres e) Sclereids

43) Virus is a latin word meaning:

a) Sweet b) Sour c)Poison d) Toxic e) Acidic

44) In the heart the thick walled chambers are located at the:

a) Apex b) Lower side c) Lateral side d) Dorsal side e) Ventral side

45) Extremely thin appendages help in:

a) Pili b) Membrane c) Flagella d) Tentacle e) Cilia

46) Colourless plastid is called:

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a) Leucoplast
b)Chromoplast c) Cyanoplast
d)Erythroplast e)Chloroplast

47) Evolution is the transformation of the form & mode of existence of an organism in such manner that the descendants from their ancestor, according to:
a) Weismann b) Lamark c) Charles
d) Zimmerman e) Darwin

48) Uterus opens to the vagina by a narrow opening bounded by muscular sphincter at its bottom known as:
a) Urinary bladder b) Cervix
c) Oviduct d)Labia
e) Vagina

49) Skeleton of marine shelled sarcodians give clues to possible deposit of:
a)Potassium b) Magnesium c) Petroleum
d) Chromium e) Calcium

50) Spinal cord comes out from the cranium though a opening known as:
a) Foramen magnum b) Lumen c) Ganglion
d) Acetabulum e) Glanoid cavity

51) Endocytosis which involve ingestion of solid material is called:
a) Solidocytosis b) Phagocytosis
c) Exocytosis d) Pinocytosis
e) Autocytosis

52) Plant like prototista called:
a) Bryophytes b) Bacteria
c) Fungi d) Tracheophytes e) Algae

53) When substrate combines the active site of an enzyme it induces a conformational changes that enable enzyme to perform:
a) Analytic function b) Metabolism c)Meosis
d) Mitosis e) Catalytic function

54) Mendel's selected following pair(s) of contrasting character:
a) One/single pair b) Seven pairs c) Three pairs
d) Four pairs e) Two pairs

55) Powerhouse of the cell is:
a) Ribosomes b) Endoplasmic reticulum e) Chloroplasts d) Mitochondria e) Chromoplasts

56) The tidal volume of lungs at rest or during normal breath is:
a) 4 liters b) 6 liters
c) 5 liters d) 3 liters
e) 2 liters

57) They detect pain and tissue damage:
a) Baroreceptors b) Chemoreceptor
e) Photoreceptor d) Nocireceptors
e) Mechanoreceptor

58) Phenotypic ratio of single trait inheritance:
a) 1:2:1 b) 3:1 c) 1:2 d) 1:3:2
e) 1:1

59) Artificial selection in breeding provides evidence for evolution is known as:
a) Domestication b) Deme
c) Gene pool d) Desire trait e)Genetic death

60) Molecules of amylose consists of unbranched chains of hundreds of glucose sub units, join together by:
a) 1-2 glycosidic linkage b) 2-3 glycosidic linkage c) 1-4 glycosidic linkage d) 1-3 glycosidic linkage e) 1-6 glycosidic linkage

61) It provide protection of coastline from storm, shelter for marine organisms, assist in carbon & nitrogen fixing:
a)Polyps b) Coral reefs
c)Medusa d) Aurelia
e) Hydra

62) Each enzyme has a dimple or groove at a specific side called:

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a) Co-factor b) Active site
c) Non-active site d) Substrate
e) Prosthetic group

63) Light reaction of photosynthesis takes place in:
3) Cristae b) Stroma
c) Cisternae d) Matrix
e) Thylakoid

64) The vacated vesicular follicle forms a fatty yellow structure on ovary called:
a) Graafian luteum b) Corpus luteum c) Ruptured ovum
d) Follicle e) Progestron

65) Achatina fulica is an African giant snail that is dangerous:
a) Agricultural pest b) For silk worm c) For paper industry
d) Apiculture pest e) Sericulture pest

66) The enzyme which is used in calvin cycle is:
a) Oxidase b) Reductase
c) Isomerase d) Rubisco
e) Pepsin

67) The flexible articulating cartilage between the joints become denatured by the deposition of calcium which makes the bone hard. The disease is:
a) Disc slip b) Rickets
c) Spondylosis d) Arthritis
e) Sciatica

68) HB molecules in each RBCs:
a) 280 million b) 150 millions
c)380 millions d) 170 millions
e) 180 millions

69) In exothermic reaction by decreasing the temperature equilibrium constant:
a. same b. sometimes decreases sometimes increase c. reaction move backward d. decrease
e. increase

70) Viscosity is measure of:
a. volume b. resistance in the flow of liquid c. fluidity of liquid d. mass e. density

71) Acetone gives :
a. Fehling solution test positive
b. silver mirror tests positive
c. Luca's test positive
d iodoform test positive
e. benedicts test

72) The equation v= RH [1/22 -1/n2] where n>2 is the equation of H spectrum for :
a. paschen series b. pfund series c. bracket series
d. Lyman series e. Balmer series

73) The equation of the first law thermodynamics q=E+W will reduce at constant volume to:
a. E+W b. Q=E+PW
c.Q=E+W d. Q=E e. Q=PGV

74) The electrolysis of dilute acid solution gives:
a. Cl2 at cathode and H2 at anode
b. H2 at cathode and Cl2 at anode
c. H2 at cathode and O2 at anode
d. H2 at anode and Na at cathode
e. H2 at anode and O2 at cathode

75) Ostwald's process is the oxidation of :
a. H2S b. NH3 c. HNO3 d. H2SO4
e. CL2

76) The dipolar moment of CO2 is zero because :
a. angular b. linear c. triatomic d. pyramidal
e. CO2 is polar

77) The two crystals NaNO3 and CaCO3 are both trigonal, they are :
a. isomers b. polymorphs c. allotropes d. enantiomers e. isomorph

78) Nylon 6,6 is a :

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a. addition polymer b. condensation polymer of hexandoic acid and 1,6 diamino hexane c. it is a ester d. condensation polymer e. it is a carboxylic acid

79) Ethene (H3C=CH3) has hybridization of C-atom :

a. dsp3 b. sp c. sp3 d. d2sp3 e. sp2

80) The kinetic molecular theory of gases explains :

a. mass number b. atoms atoms no c. the behaviors of gases d. electrons e. neutrons

81) 2-chloropropane is heated with Na metal product is 2,3 dimethyl butane reaction is :

a. dehydrohalogenation b. wurtz reaction c. dehydration d. catalytic hydrogenation e. dehalogenation

82) At equilibrium in a reversible reaction:

a. the rate of backward reaction >rate of forward reaction

b. the rate of forward reaction > rate of backward reaction

c. the rate of forward reaction is same at backward reaction

d. the concentration of reactants and products becomes constant and rate of forward reaction is same as the rate of backward reaction e. the concentration of reactant and product becomes constant

83) Rate of reaction increases by addition of catalyst because:

a. catalyst reacts with reactant

b. catalyst reacts with product

c. energy of activation increases

d. energy of activation increases

e. energy of activation remains same

84) Reaction $H_2+Cl_2 \rightarrow 2HCl$ is a :

a. second order b. first order c. half order d. third order e. zero order

85) The comparision of ratio of diffusion of H2 and O2 is in the ratio of :

a. 2:1 b. 1:4 c. 3:2 d. 4:1 e. 1:2

86) Grignard reagent, e.g. methyl magnesium halide reacting with methyl halide product is :

a. N-propyl chloride b. butane c. isopropyl chloride d. methane e. ethene

87) HOOC-COOH has IUPAC name :

a. acetic acid b. oxalic acid c. diethanoic acid d. ethandioic acid e. benzoic acid

88) Pulverized CsCu3 reacts faster because of the increase in:

a. pressure b. surface area c. volume d. surface tension e. temperature

89) Oxidation of K2MnO4 by Cl2 Is to manufacture :

a. K3CrO4 b. MnO3 c. K2Cr2O4 d. KMnO4 e. O2

90) Aldehydes reacts with:

a. ketones are more easily oxidized

b. it is always difficult to oxidize aldehyde

c. mild oxidizing agents and strong oxidizing agents

d. only strong oxidizing agent

e. only mild oxidizing agents

91) Benzene is subjected to alkylation, nitration and oxidation the product is :

a. o & p nitro toluene b. m-nitro benzoic acid c. m-nitro toluene d. o&p nitro benzoic acid e. onitro benzoic acid

92) In castner kellners cell :

a. NaOH+H2 are produced at cathode

b. Cl2 is produced at anode

c. NaOH +H2 at anode

d. NaOH+H2 are produced at cathode and Cl2 is produced at anode

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e. Cl2 is produced at anode e. Cl2 is produced at cathode

93) Bakelite is a polymer :

a. two glucose molecules

b. formaldehyde and acetone

c. two amino acid molecules

d. formaldehyde and phenol

e. acetone and carboxylic acid

94) Phenol when subjected to nitration the product is:

a. o-nitro phenol b. p-nitro phenol

c. o nitro phenol & m-nitro phen d.o-nitro phenol and pnitro phenol

e. m-nitro phenol

95) Heating a mixture of chromite ore (FeOCr2O3) & K2CO3 in air to manufacture:

a. K2MnO4 b. K2Cr2O3 c. KMnO4 d. KI e. K2CrO4

96) A strong electrolyte is one that —— completely in solution:

a. reacts b. decomposes c. disappears d. ionizes e. never ionizes

97) Cation & anion formed which combines to form crystal lattice :

a. only cation is isoelectric to near noble gases b. cation and anion both are isoelectronic to nearer noble gases

c. cation & anion both are isoelectronic to nearest noble gas & number of electrons is lost metal & no of electrons lost by non metal d. only anion is isoelectronic to nearest noble gases

e. no of electrons lost by metal &no of electrons gained by non metal is equal

98) Secondary alcohols have :

a. one βC b. three βC c. no βC d. two βC e. four βC

99) Small diffusion in solids is due to :

a. very fast motion of its particle

b. vibrational motion of its particles

c. rotational motion of its particles

d. translational motion of its particles

e. diffusion in solids is not present

100) Gases deviate from ideal behavior more :

a. gases do not deviate from ideal behavior

b. both temperature and pressure low

c. both temperature and pressure high

d. at high temp and pressure high

e. at low pressure and high temperature

101) 3.0 gram of C on combustion gives C+O2= CO2 :

a. 5.5g of CO2 b. 22g of CO2 c. 6.02×10^23 molecules of CO2 d. 3.01 ×10^23 molecules of CO2 e. 1.505 ×10^23 molecules of CO2

102) Alkyl halide on reaction with NaOH, product is an alkane. Mechanics of reaction is :

a. SN b. elimination c. decomposition d. electrophilic substitution reaction e. addition

103) The first organic compound manufactured from inorganic compound was :

a. ethyl acetate b. urea c. acetic acid d. methane e. acetone

104) The ionization energy increases along a period in periodic table due to increase in:

a. no of positrons b. no of neutrons c. nuclear charge

d. no of electrons e. atomic size

105) Reaction taking place in one step or in several steps have same ΔH it is statement of:

a. lusaac's law of combining volumes

b. Hess's law c. Boyle's law d. Charles's law

e. Avogadro's law

106) Oxidation of primary alcohol gives:

a. aldehydes b. ketone c. alkane d. alkene e. alkaloids

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107) Primary alcohols when reacts with halogen acids the product is:
a. secondary alcohols b. tertiary alcohols c. primary alcohols d. primary alkyl halide e. secondary alkyl halide
108) Acetophenone is an:
a. aliphatic ketone b. phenols
c. carboxylic acids d. aldehydes
e. aromatic ketones
109) SN1 mechanism is a:
a. formation of carbocation in first step
b. single step mechanism
c. two step mechanism
d. attack of nucleophile on carbocation is slow step
e. primary alkyl halide follows Sn1 mechanism
110) Hexanoic acid is IUPAC name of :
a. adipic acid b. propanoic acid
c. formic acid d. caproic acid e. butyric acid
111) The options having all the properties of γ rays :
a. neutron highly penetrating, velocity is equals to that of light , remains undeflected in magnetic field
b. most penetrating, velocity is equals to that of light no deflection in magnetic field
c. least penetrating, velocity 1/10th of light deflected in magnetic field helium nucleus
d. proton least penetrating, deflects like negative particle in magnetic field
e. most penetrating power, velocity less than that of light deflects in magnetic field they are electrons
112) The solubility product Ksp predicts whether :
a. diffusion b. solubility c. precipitation will take place or no d. boiling point
e. melting point
113) In protein amino acid molecules are linked by:
a. glycosidic linkage b. glycosidic linkage c. peptide linkage
d. ethane linkage e. ester linkage
114) Ratio of atomic mass of hydrogen to atomic mass of carbon is:
a. one sixth b. one twelfth c. half d. double e. equal
115) Reforming :
a. increases knocking of internal combustion engine
b. increases octane number
c. more is knocking better is the fuel
d. to produce a fuel of less octane number
e. decreases the octane number
116) Addition product of formaldehyde and ethyl alcohols is :
a. aldehyde b. acetal c. carboxylic acid
d. acetone e. ketone
117) Down's cell is for the :
a. metallurgy of Na b. metallurgy of Mg c. metallurgy of Al
d. metallurgy of Fe e. metallurgy of Ca
118) Falling drop of liquid is spherical due to;
a.evaporation on exposure b. surface tension c. atomic atomic structure
d. viscosity e. vapors pressure
119) Catalytic hydrogenation of oil gives;
a. alkene b. alkyne c. soap d. alkyl halides e. fat
120) Benzyl alcohol C6H5CH2OH is a;
a. aromatic alcohol b. phenol
c. aromatic ketone d. aliphatic alcohol e. alicyclic alcohols
121) Ammonia Solvay process is an industry for manufacture of;
a. HNO3 b. HCL c. soda ash (Na2CO3) d. H2SO4 e. CaCO3
122) The quantum number allows following orbitals in M shell.

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a. s-orbital b. s,p,d&f-orbitals c. s&p orbitals d. all options are correct e. s,p&d orbitals

123. What different resistors are connected across the terminals of battery:
a. both emf and P.D becomes zero
b. both emf and terminal P.D changes
c. the emf changes but terminal P.D ……
d. both emf and terminal P.D remains the same e. the emf remains the but P.D changes
124) The minimum energy of pair production is :
a. 1.02 V b. 1.02eV c. 1.02 MeV d. 1.02 GeV e. 1.02 KeV
125) The mathematical equation representing faradays law is :
a. $E=N\Delta I/\Delta t$ b. $E=-N\Delta\Theta/\Delta t$ c. $E=N\Delta I/\Delta t$ d. $E=\Delta\Theta/\Delta t$ e. $E=-(\Delta\Theta)\ \Delta t$
126) The cyclist moves on circular track skids because :
a. the limiting friction is perpendicular to centripetal force
b. the limiting friction is greater than required centripetal force
c. limiting friction is opposite to centripetal force
d. the limiting friction is equals to required centripetal force
e. the limiting friction is less than the required centripetal force
127) A body is dropped from certain height and falls freely its velocity after 5 seconds will be :
a. 94 m/s b. 9.4 m/s c. 0.49 m/s d. 4.9 m/s e. 49 m/s
128) Two transverse travelling waves are presented by these equations yt=A•sin(kx-ek) and y2= A•sin(kx-ax-w) these two waves differ in :
a. amplitude b. frequency c. wavelength d. phase e. direction of propagation
129) In equation $E=-\Delta V/\Delta r$ negative sign indicates that :
a. E points towards increasing V
b. E points decreasing V
c. E is perpendicular to V
d. E is parallel to ΔV
e. E is opposite to ΔV
130) The path of neutron moving perpendicular to magnetic field is :
a. curve b. straight line c. circle d. ellipse e. parabola
131) A battery of emf E has an internal resistance "r". if current "I" is drawn then terminal potential Vt is given by :
a. Vt=E+Ir b. Vt=E-Ir c. Vt=E/I-r d.Vt=Er
e. Vt=E/I+r
132) The instantaneous emf generated by alternating current generator is written as :
a. $E=BINA\sin\omega t$ b. $E=BNA\sin\omega t$ c. $E=BINA$
d. $E=E0/BINA$ e.$E=BINA\omega$
133) A bullet is fired horizontally with 20m/s in the absence of air friction its horizontal velocity after 2 seconds :
a. 10 m/s b. 05 m/s c. 60 m/s d. 40 m/s
e. 20 m/s
134) If Rydberg constant for hydrogen is RH then wavelength of line in Lyman series is :
a. 4RH /3 b. RH/4 c. 4/RH d. 4/3RH e. RH /2
135) The ratio of molar specific of a gas at constant pressure to its molar specific heat at constant volume is :
a. equals to minus b. equals to zero c. greater than one d. less than one e. equal to one
136) If mass and speed of moving object is double, the K.E will be :

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a. eight times b. four times c. doubled d. sixteen times e. six times

137) The electric field intensity b/w two uniformly, appositively charged parallel plates is a :

138) The box of mass M=6kg slides with speed V=4m/s across a frictionless floor. It suddenly explodes into two pieces . one piece with mass M1=2kg moves in the same direction with speed v1=5m/s. the velocity of second place is :

a. 6m/s b. 3m/s c. 1m/s d. 2m/s e. 4m/s

139) The intensity level(sound level) of faintest audible intensity is :

a. 160 decibel b. zero decibel c. 140 decibel d. 120 decibel e. 60 decibel

140) If 2 uF capacitors are connected in series . their equivalent capacitance will be :

a. 8 uF b. 0.75 uF c. 1.33 uF d. 6uF e. 2uF

141) The experimental proof of particle nature of light is :

a. pair production b. photoelectric effect c. davison-germer experiment d. de-Broglie concept e. Compton's effect

142) The electrical energy dissipated as heat in a resistor is :

a. V2R b. V2Rt c. I2R d. I2Rt e. IR

143) One kilowatt hour is equals to :

a. 36×10^{6} J b. 36×10^{-6} J c. 3.6×10^{5} J d. 36×10^{-5} J e. 36×10^{5} J

144) Unit joule per coulomb Is equals to :

a. volt b. ampere c. tesla d. henry e. weber

145) According to law of radioactive decay , the of disintegration is directly proportional to:

a. pressure b. density of substance c. volume of substance d. number of atoms present e. temperature

146) The eddy current loss in a transformer can be reduced by :

a. using thick copper coils b. using thick non magnetic core c. using iron core d. using laminating iron core e. temperature

147) The absolute gravitational P.E of an object of mass m on earth's surface is given by:

a. U=-GmM/R b. U=GmM/R c. $U=GmM/Re^{2}$ d. $U=GMe/Re^{2}$ e. U=-GMe/Re

148) The speed of a photon in free space is:

a. $1/\sqrt{u0e0}$ b. ½ uoc0 c. $1/u0e0$ d. $\sqrt{}$ u0e0 e. u0e0

149) A charge q is moving with velocity v, parallel to magnetic field B. the magnetic force F on charge q is:

a. $F=qVB\sin\Theta$ b. F=qVB c. F=-qVB d. F=0 e. F=qB/v

150) Resistors of 5Ω and 10Ω are connected parallel if the P.D across 5Ωis 20volt then current through 10Ω will be :

a. 0.5 ampere b. 20 ampere c. 4 ampere d. 10 ampere e. 2 ampere

151) A helicopter of weight 'W' is ascending with zero acceleration. The upward force F on it is:

A. F > W B. F = $\sqrt{}$W C. F = 0 D. F = W E. F < W

152) A sound wave travel from one medium to another which of following characteristics of sound remain unchanged:

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A. Amplitude B. Velocity C. Pitch D. Quality
E. Wavelength

153) For efficient transfer of electricity through cables over the long distance :
A. Both the current and voltage must be low B. The current must be low and voltage high
C. Both the current and voltage must be high D. The voltage must be zero and current low
E. The voltage must be low and current high

154) When a body moves with uniform velocity then it's :
A. Average velocity is zero
B. Instantaneous velocity is greater than average velocity
C. Instantaneous velocity is zero
D. Instantaneous velocity is equal to average velocity
E. Instantaneous velocity is less than average velocity

155) Every particle of rotating disc moves with : A.
Constant linear velocity
B. Constant angular velocity
C. Constant tangential velocity
D. Constant linear displacement
E. Constant linear momentum

156) A proton of charge $1.6x10^{-19}$ C is moved through potential difference of 20 V. The energy gained by proton is :
A. $-32x10^{-19}$ J B. $32x10^{19}$ J C. $-32x10^{19}$ J D. $32x10^{19}$ J E. Zero

157) Distance between two consecutive antinodes in standing wave is :
A. $\lambda/4$ B. λ C. 2λ D. $\lambda/2$ E. $3/2\ \lambda$

158) A conducting wire of resistivity 'ρ' is cut into two equal parts. The resistivity of each part will be :
A. The same B. One fourth C. $\sqrt{2}$ times D. Halved E. Doubled

159) If the giving spring of spring constant K is cut into three identical segments the spring constant of each segment is :
A. 1K B. K/2 C. K/3 D. 3K
E. 4K

160) Four cells of emf three volts are connected in series to form a battery. Net emf of combination is:
A. 12 V B. 1.33 V C. 9 V D. 7 V
E. 3 V

161) The rate of change of linear momentum of body is equal to :
A. Net force B. Net displacement
C. Net torque D. Net inertia
E. Net velocity

162) The most appropriate mathematical expression representing simple harmonic motion is :
A. $a \propto x$ B. $a \propto -x$ C. $a = x$ D. $a = kx$
E. $a = -x$

163. A body is moving in a circle with constant speed with radius of circle r and time period 'T' the centripetal acceleration is given by:
A. $a_c = 4\pi r^2/T^2$
B. $a_c = 4\pi^2 r/T^2$
C. $a_c = 4\pi r/T$
D. $a_c = 4\pi^2 r^2/T$
E. $a_c = 4\pi r^2/T$

164) The kinetic energy of an electron in first Bohr radius of H-atom is :
A. 13.6 J B. 13.6eV C. -13.6eV D. 13.6keV
E. 13.6 kJ

165) Lenz's law is a form of the :
A. Law of conservation of momentum
B. Law of conservation of charge
C. Law of conservation of angular momentum
D. Law of conservation of energy
E. Law of conservation of mass

166) The rate of doing work is zero when angle between Force and velocity is :

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A. $\theta = 60^o$ B. $\theta = 180^o$ C. $\theta = 0^o$ D. $\theta = 45^o$ E. $\theta = 90^o$

167. Half life of radioactive sample is given by :

168) Two similar point charges each of 1C placed in free space 1m apart, repel each other with force of :

A. 8.85×10^{-12} N B. 1.602×10^{-19} N C. 1N D. 6.67×10^{11} N E. 9×10^9 N

169) In nuclear reaction equation. the missing particle is :

A. Proton B. Alpha particle C. Electron D. Positron E. Neutron

170) The potential difference applied across a diode to reduce depletion layer is :

A. Reverse biasing B. Forward biasing C. Biasing voltage D. Junction potential E. Potential barrier.

171) A charged particle of mass m and charge q is projected in magnetic field of induction B at an angle θ. The radius of curvature of curved path is given by :

A. $r = mv/qB$ B. $r = mv/qB\sin\theta$ C. $r = mv\cos\theta/qB$ D. $r = mv\sin\theta/qB$ E. $r = mvB/q$

172) Specific heat capacity of a gas is measured in :

A. $J.kg.K^{-1}$ B. $J.kg^2.K$ C. $J.kg^{-1}.K^{-1}$ D. $J.kg^{-1}.K$ E. J.kg.K

173) First law of thermodynamics is mathematical expressed as :

A. $\Delta W = \Delta Q + \Delta U$ B. $\Delta U = \Delta W - \Delta Q$ C. $\Delta Q = \Delta W - \Delta U$ D. $\Delta Q = \Delta U - \Delta W$ E. $\Delta U = \Delta Q - \Delta W$

174) The angular momentum of particle changes from 0 to 720J.s in 4s. The magnitude of torque is :

A. 2880 Nm B. 180 Nm C. 360 Nm D. 1440 Nm E. 90 Nm

175) An object moving faster than speed of sound. The object is called :

A. Sonic B. Supersonic C. Hypersonic D. Ultrasonic E. Infrasonic

176) If a slab dielectric constant Er =2 is inserted between , the parallel plate of a capacitors 20uF capacitor . its capacitance will :

a. remains the same b. be four times c. be doubled d. be halved e. becomes zero

177.Fill in with appropriate preposition; we saw a cricket match _____ the school stadium. :

a. in b. over c. of d. at e. on

178. I _____ an appointment with the dentist soon :

a. making b. had made c. make d. made e. will make

179. due to his negligence, he failed in the exam :

a. he failed b. in the examination c. his negligence d. no error e. due to his negligence

180. which sentences uses capital letters correctly

a. I will be moving to China
b. i will be moving to China
c. I will be moving to China
d. I will be moving to China
e. I will be moving to China

181. Fill in with appropriate article; To climb _____ tree is not to climb _____ mountain :

a. a/a b. an/an c. a/an d. the/an e. an/a

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182. Error correction; I've lived in Barcelona since six year :
a. has six years b. been six years c. at six years
d. has been six years e. for six years
183. Spot the correct spelling :
a. coomitte b. commitee c. committey d. committee e. comittee
184) To _____ , Anne was on time for her math class :
a. everybody's surprise b. everybody surprise c. everybody's surprise d. ever body surprise e. every-body surprise
185. Find the error;
a. she shows us five different shades of blue paint b. She was approached, but she declined the offer
c. There are seven floors in this building
d. this is the best birthday party I have ever had e. when I go the museum
186. Figure out the error; scientist now hope that cloning can successfully be conducted in human beings in the near future.
a. can successfully be conducted
b. scientist now hope that cloning
c. in near future
d. human beings in near future e. no error
187. fill in with appropriate; ____ apple ____ day keeps____ doctor away :
a. the/a/a b. an/a/an c. an/the/the
d. a/a/the e. an/a/the
188. fill in with appropriate preposition; she climbed ____ the ladder to paint the wall :
a. to b. up c. of d. on
e. in
189. Charles____ his father in the shop until school _____ ?
a. is helping, starts b. had help, started c. helped , was starting
d. has helped, is starting e. was helping, will start
190. fill in suitable verb; the teacher_____completed this chapter :
a. have b. is c. are d. has
e. they
191. one must consider about any matter before_____ it:
a. decided b. decides c. decode d. was decide
e. deciding

Read the passage to answer question 192-194. But man is not destined to vanish. He can be killed but he can not be destroyed, because his soul is deathless and his spirit is irrepressible. Therefore, though the situation seems dark in the context of the confrontation between the super powers , the silver lining is provided by amazing phenomena that the very nations which have spent incalculable resources and energy for the production of deadly weapons are desperately trying to find out how they might never be used . they threaten each other and go to the brink, but before the total hour arrives they withdraw from the brick.
192. a suitable tittle for the above passage is :
a. cost of living
b. man's desire to survive inhibits use of deadly weapons
c. threats and intimidation between super powers
d. destruction of mankind is inevitable
e. mounting cost of modern weapons.
193. the main point of the authors view is that;
a. Man has come here to live
b. man's soul and spirit cannot be destroyed
c. man's destiny is not fully clear or

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visible
d. man's soul and spirit are immortal
e. human society will survive despite the serious threat of total annihilations

194. "Irrepressible" in the second line means
a. harmful b. incompatible c. strong
d. oppressive e. unrestrainable

195) P,Q,R,S, and T are five people in the family . if P is the daughter of Q, Q is the son of R, R is the father of S , while T is the daughter of P , then which of the following is true ;
a. R is the uncle of P b. R is the grandfather of P c. P and R are sisters
d. Q is the daughter of S e. P is the sister of T

196) which would be the next the series?
7C 10F 14J 19O
a. 23Q b. 26N c. 27X d. 25U e. 24P

197) although most of the lucrative jobs in Pakistan will require a college degree. Many of the jobs require knowledge other than in colleges or universities . for workers in those jobs, good basic skills in reading , communications and mathematics plays an important role in getting a job & developing a career . from the given information above what can be validity can be concluded in today's economy.
a. extra knowledge is added in benefit for career development
b. people should train their children for technical work
c. skills in reading , communication & mathematics play an important role in developing a successful career.

198) after seven years of suggestion from the environment department , the Sindh legislation has finally passed an antipollution ordinance . from a reading of the language , the legislation promises to be one of the most effective bills in the history of province , which of the following can be deducted :
a. pollution in not now a problem in the country
b. to reduce pollution, the legislation must now be enforced
c. the pollution will be reduced in the country
d. the pollution problem will be eliminated in the country
e. the problem will be reduced in the country

199) after the declaration of company election , the elected chairperson gave his message to the workers that "the responsibility of the management is to serve not to dominate the workers. Which of the following one basic assumption underlying chairperson's statement ?
a. the management is capable of dominating the workers
b. one must be decessive when facing company's union for the first time
c. domination over workers by the management is virtue
d. the management preceding the company administration had been irresponsible
e. the manager chooses to serve rather than dominate the workers

200) scientists recently published that influenza spreads around the world more efficiently in the modern era due to commercial air travel , flu symptoms are severe enough that the sick would likely cancel the air travel, but an infected person can travel across the globe before the first sign appears . further, if symptoms develop while someone is on plane , the patients cough can spreads the virus easily in the enclosed and closely packed environment . which of the following would best minimize the role of

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air travel can play in the spread of influenza during a pandemic ?
a. refusing to allow children, the elderly or others who are specially vulnerable to flu to travel by air during pandemic
b. requiring all air travelers to wash their hands before boarding a plane
c. requiring air travelers to receive flu vaccination far enough in advance of the trip to provide flowing through the filters
d. installing air filters system in the plane to kill any flu virus particles flowing through the filters
e. conducting medical examination during the boarding process and prohibiting passengers with flu symptoms from flying

End The
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2000-2023

Dow University of Health Sciences Karachi

Examinations Department

Medical Universities of Sindh

MDCAT – Test 2022

Held on 13th November 2022

Code No: B Green Answers KEY

1	E	51	B	101	E	151	D
2	E	52	E	102	A	152	C
3	C	53	E	103	B	153	B
4	B	54	B	104	C	154	D
5	B	55	D	105	B	155	B
6	D	56	E	106	A	156	B
7	B	57	D	107	D	157	D
8	C	58	B	108	E	158	A
9	D	59	A	109	C	159	D
10	E	60	C	110	D	160	A
11	A	61	B	111	B	161	A
12	C	62	B	112	C	162	B
13	A	63	E	113	C	163	B
14	E	64	B	114	B	164	B
15	D	65	A	115	B	165	D
16	C	66	D	116	B	166	E
17	E	67	D	117	A	167	E
18	B	68	A	118	B	168	E
19	C	69	E	119	E	169	A
20	A	70	B	120	A	170	B
21	A	71	A	121	C	171	D
22	E	72	E	122	E	172	C
23	E	73	D	123	E	173	E
24	A	74	B	124	C	174	B
25	E	75	B	125	B	175	B
26	B	76	B	126	E	176	C
27	C	77	E	127	E	177	D
28	A	78	B	128	D	178	E
29	B	79	E	129	B	179	D
30	C	80	C	130	B	180	D
31	D	81	B	131	B	181	A
32	D	82	D	132	B	182	E
33	C	83	C	133	E	183	D
34	A	84	E	134	D	184	C
35	B	85	D	135	C	185	E
36	B	86	E	136	A	186	B
37	B	87	D	137	D	187	E
38	D	88	B	138	E	188	B
39	B	89	D	139	B	189	A
40	E	90	C	140	C	190	D
41	C	91	D	141	B	191	E
42	C	92	D	142	D	192	B
43	C	93	D	143	E	193	E
44	A	94	D	144	A	194	E
45	A	95	E	145	D	195	B
46	A	96	D	146	D	196	D
47	E	97	C	147	A	197	C
48	B	98	D	148	A	198	B
49	C	99	B	149	D	199	A
50	A	100	D	150	E	200	C

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